Business Intelligence

Chapter 4:

Data Mining for Business Intelligence

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Learning Objectives

- Define data mining as an enabling technology for business intelligence
- Understand the objectives and benefits of business analytics and data mining
- Recognize the wide range of applications of data mining
- Learn the standardized data mining processes
  - CRISP-DM
  - SEMMA
  - KDD

Learning Objectives

- Understand the steps involved in data preprocessing for data mining
- Learn different methods and algorithms of data mining
- Build awareness of the existing data mining software tools
  - Commercial versus free/open source
- Understand the pitfalls and myths of data mining

Opening Vignette…

"Data Mining Goes to Hollywood!"

- Decision situation
- Problem
- Proposed solution
- Results
- Answer & discuss the case questions

Opening Vignette: Data Mining Goes to Hollywood!

Class No.	1	2	3	4	5	6	7	8	9
Range (in $Millions)	< 1 (Flop)	> 1 < 10	> 10 < 20	> 20 < 40	> 40 < 65	> 65 < 100	> 100 < 150	> 150 < 200	> 200 (Blockbuster)

Dependent Variable

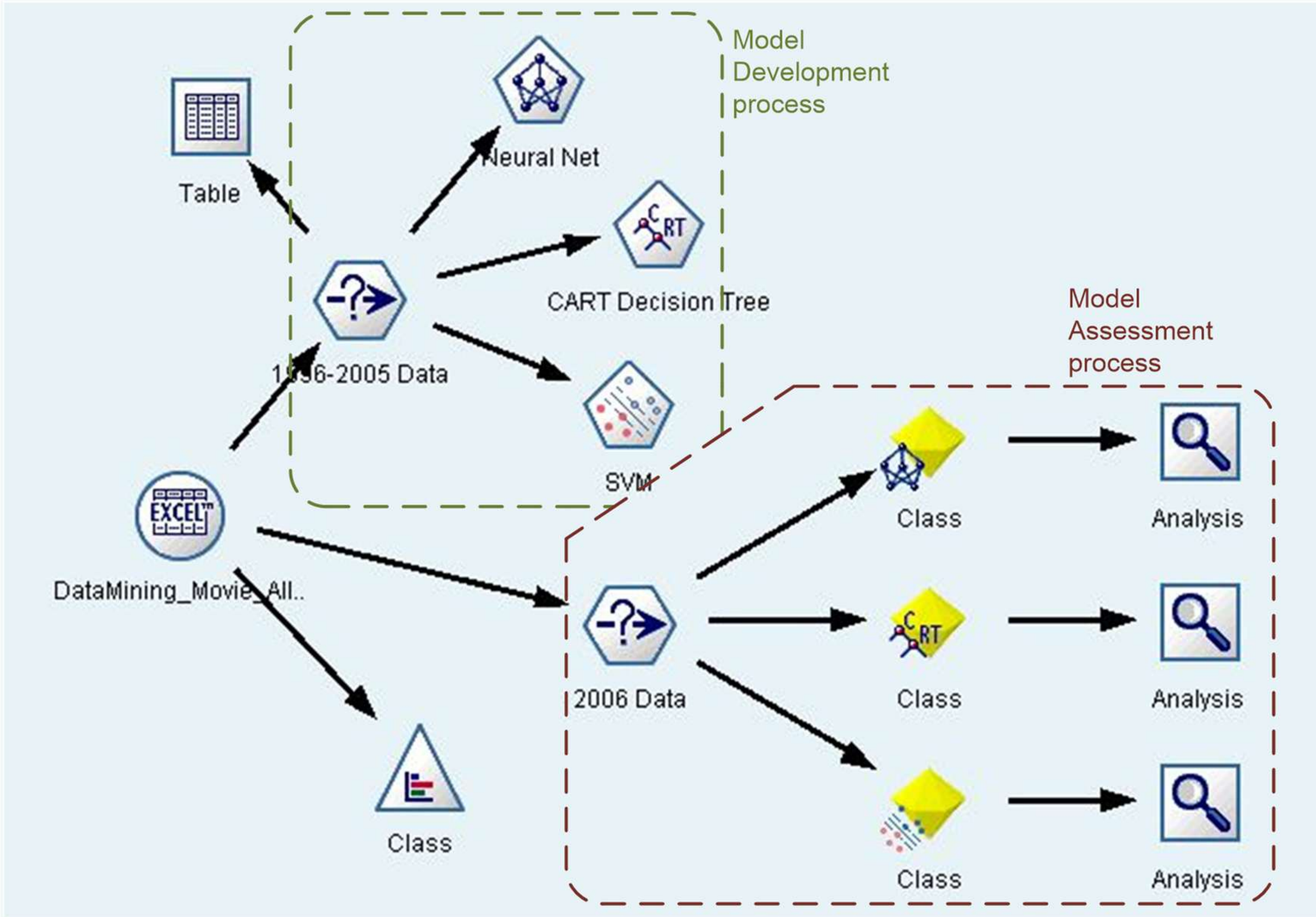
Independent Variables

Independent Variable	Number of Values	Possible Values
MPAA Rating	5	G, PG, PG-13, R, NR
Competition	3	High, Medium, Low
Star value	3	High, Medium, Low
Genre	10	Sci-Fi, Historic Epic Drama, Modern Drama, Politically Related, Thriller, Horror, Comedy, Cartoon, Action, Documentary
Special effects	3	High, Medium, Low
Sequel	1	Yes, No
Number of screens	1	Positive integer

A Typical Classification Problem

Opening Vignette:
Data Mining Goes to Hollywood!

The DM Process Map in IBM SPSS Modeler



Opening Vignette:
Data Mining Goes to Hollywood!

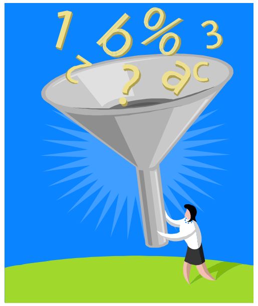
	Prediction Models					
	Individual Models			Ensemble Models		
Performance Measure	SVM	ANN	C&RT	Random Forest	Boosted Tree	Fusion (Average)
Count (*Bingo*)	192	182	140	189	187	**194**
Count (*1-Away*)	104	120	126	121	104	**120**
Accuracy (*% Bingo*)	55.49%	52.60%	40.46%	54.62%	54.05%	**56.07%**
Accuracy (*% 1-Away*)	85.55%	87.28%	76.88%	89.60%	84.10%	**90.75%**
Standard deviation	0.93	0.87	1.05	0.76	0.84	**0.63**

* *Training set: 1998 – 2005 movies; Test set: 2006 movies*

Data Mining Concepts and Definitions
Why Data Mining?

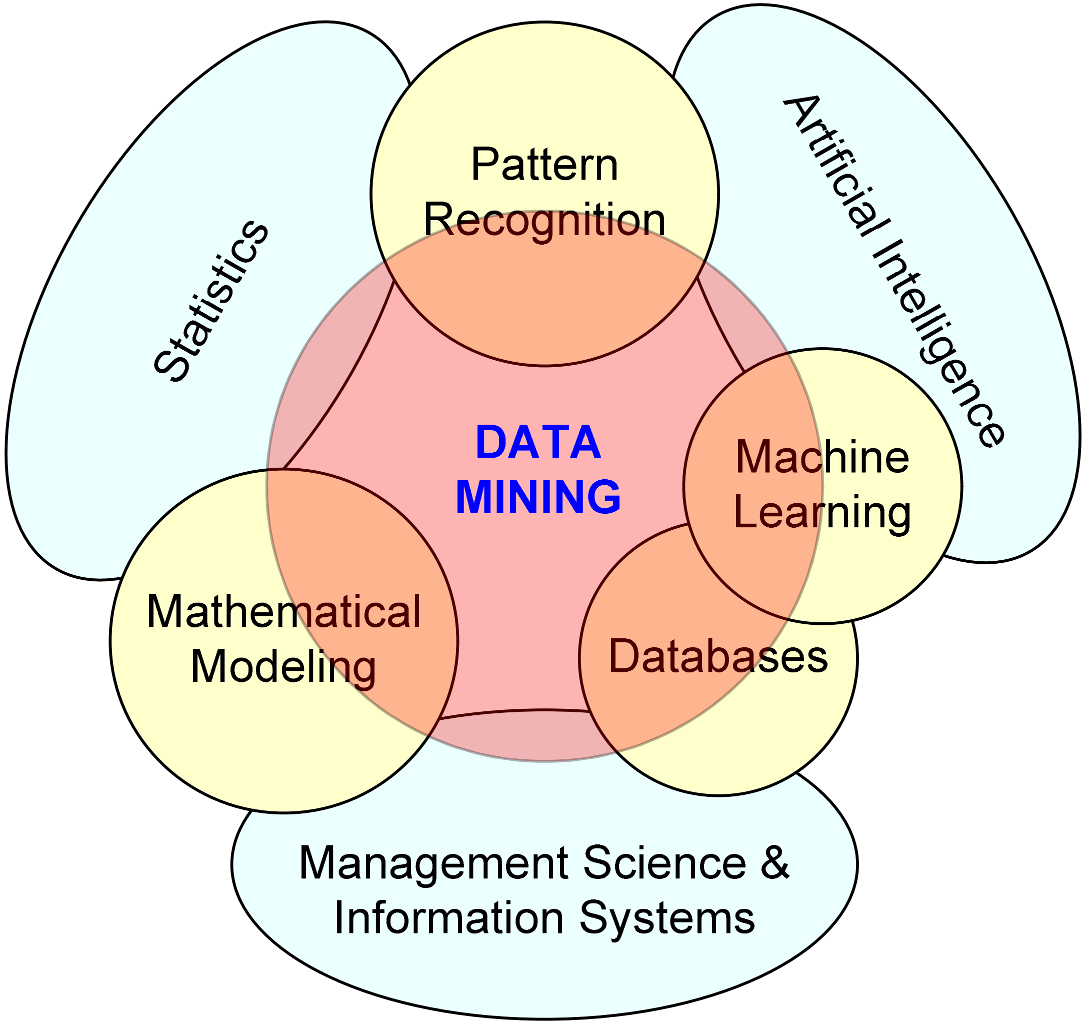
- More intense competition at the global scale
- Recognition of the value in data sources
- Availability of quality data on customers, vendors, transactions, Web, etc.
- Consolidation and integration of data repositories into data warehouses
- The exponential increase in data processing and storage capabilities; and decrease in cost
- Movement toward conversion of information resources into nonphysical form

Definition of Data Mining



- The nontrivial process of identifying valid, novel, potentially useful, and ultimately understandable patterns in data stored in structured databases
  - Fayyad et al., (1996)
- <u>Keywords in this definition</u>: Process, nontrivial, valid, novel, potentially useful, understandable
- Data mining: a misnomer?
- Other names: knowledge extraction, pattern analysis, knowledge discovery, information harvesting, pattern searching, data dredging

Data Mining at the Intersection of Many Disciplines

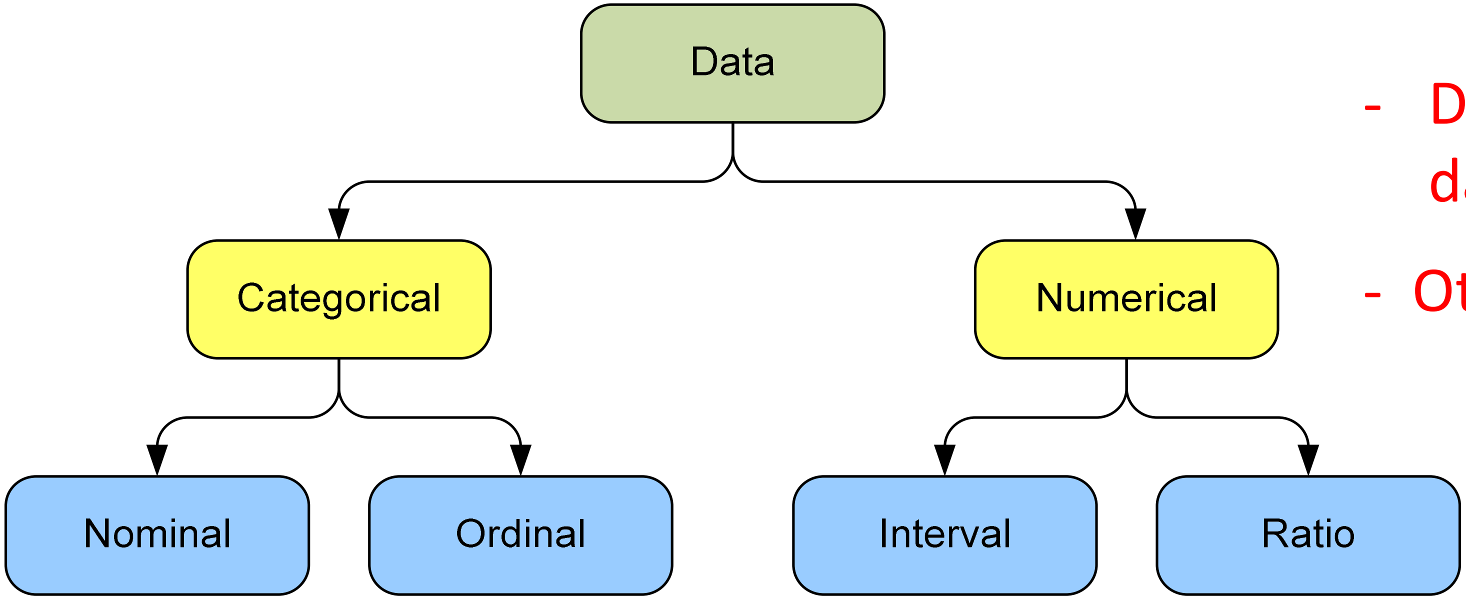


Data Mining Characteristics/Objectives

- Source of data for DM is often a consolidated data warehouse (not always!).
- DM environment is usually a client-server or a Web-based information systems architecture.
- Data is the most critical ingredient for DM which may include soft/unstructured data.
- The miner is often an end user.
- Striking it rich requires creative thinking.
- Data mining tools' capabilities and ease of use are essential (Web, Parallel processing, etc.).

Data in Data Mining

- Data: a collection of facts usually obtained as the result of experiences, observations, or experiments
- Data may consist of numbers, words, and images
- Data: lowest level of abstraction (from which information and knowledge are derived)



- DM with different data types?
- Other data types?

What Does DM Do? How Does it Work?

- DM extracts patterns from data
  - Pattern?
    A mathematical (numeric and/or symbolic) relationship among data items
- Types of patterns
  - Association
  - Prediction
  - Cluster (segmentation)
  - Sequential (or time series) relationships

A Taxonomy for Data Mining Tasks

Data Mining	Learning Method	Popular Algorithms
Prediction	Supervised	Classification and Regression Trees, ANN, SVM, Genetic Algorithms
Classification	Supervised	Decision trees, ANN/MLP, SVM, Rough sets, Genetic Algorithms
Regression	Supervised	Linear/Nonlinear Regression, Regression trees, ANN/MLP, SVM
Association	Unsupervised	Apriory, OneR, ZeroR, Eclat
Link analysis	Unsupervised	Expectation Maximization, Apriory Algorithm, Graph-based Matching
Sequence analysis	Unsupervised	Apriory Algorithm, FP-Growth technique
Clustering	Unsupervised	K-means, ANN/SOM
Outlier analysis	Unsupervised	K-means, Expectation Maximization (EM)

Other Data Mining Tasks

- These are in addition to the primary DM tasks (prediction, association, clustering)
  - Time-series forecasting
    - Part of sequence or link analysis?
  - Visualization
    - Another data mining task?

- Types of DM
    - Hypothesis-driven data mining
    - Discovery-driven data mining

Data Mining Applications

- Customer Relationship Management
  - Maximize return on marketing campaigns
  - Improve customer retention (churn analysis)
  - Maximize customer value (cross- or up-selling)
  - Identify and treat most valued customers
- Banking & Other Financial
  - Automate the loan application process
  - Detecting fraudulent transactions
  - Maximize customer value (cross- and up-selling)
  - Optimizing cash reserves with forecasting

Data Mining Applications (cont.)

- Retailing and Logistics
  - Optimize inventory levels at different locations
  - Improve the store layout and sales promotions
  - Optimize logistics by predicting seasonal effects
  - Minimize losses due to limited shelf life
- Manufacturing and Maintenance
  - Predict/prevent machinery failures
  - Identify anomalies in production systems to optimize manufacturing capacity
  - Discover novel patterns to improve product quality

Data Mining Applications (cont.)

- Brokerage and Securities Trading
  - Predict changes on certain bond prices
  - Forecast the direction of stock fluctuations
  - Assess the effect of events on market movements
  - Identify and prevent fraudulent activities in trading
- Insurance
  - Forecast claim costs for better business planning
  - Determine optimal rate plans
  - Optimize marketing to specific customers
  - Identify and prevent fraudulent claim activities

Data Mining Applications (cont.)

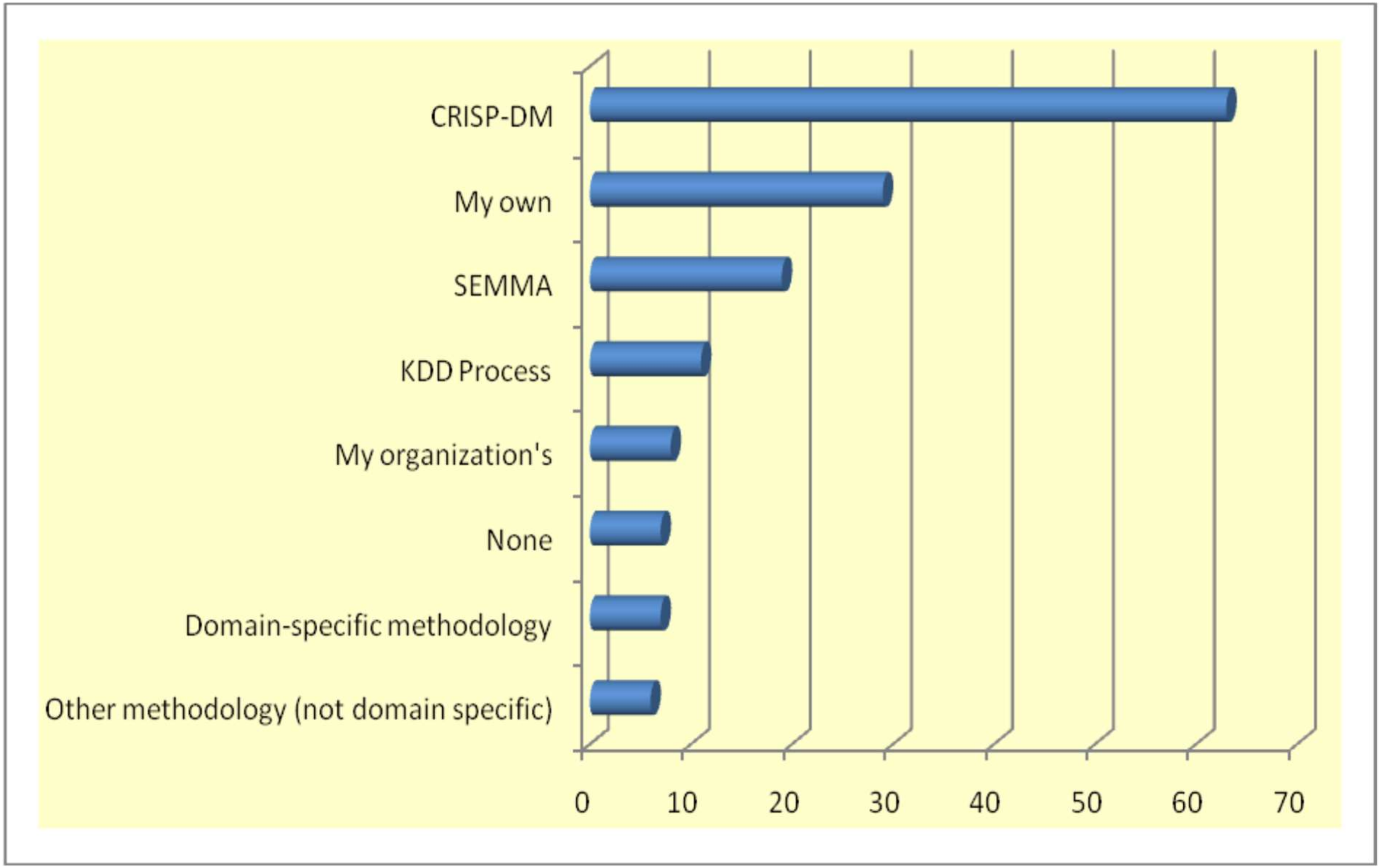
- Computer hardware and software
- Science and engineering
- Government and defense
- Homeland security and law enforcement
- Travel industry
- Healthcare
- Medicine
- Entertainment industry
- Sports
- Etc.

Highly popular application areas for data mining

Data Mining Process

- A manifestation of best practices
- A systematic way to conduct DM projects
- Different groups have different versions
- Most common standard processes:
  - CRISP-DM (Cross-Industry Standard Process for Data Mining)
  - SEMMA (Sample, Explore, Modify, Model, and Assess)
  - KDD (Knowledge Discovery in Databases)

Data Mining Process



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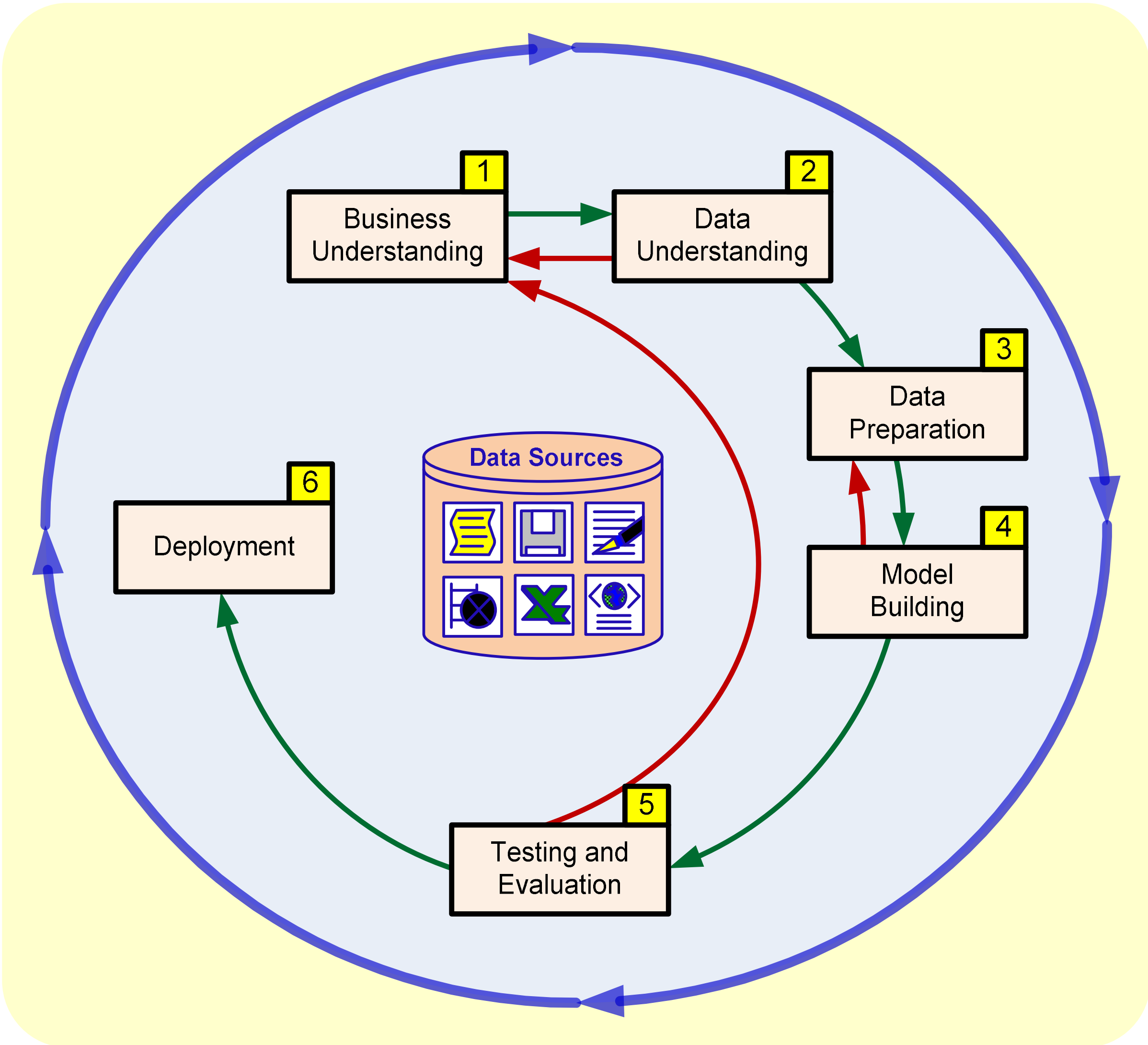
Data Mining Process: CRISP-DM



Data Mining Process: CRISP-DM

Step 1: Business Understanding

Step 2: Data Understanding

Step 3: Data Preparation (!)

Step 4: Model Building

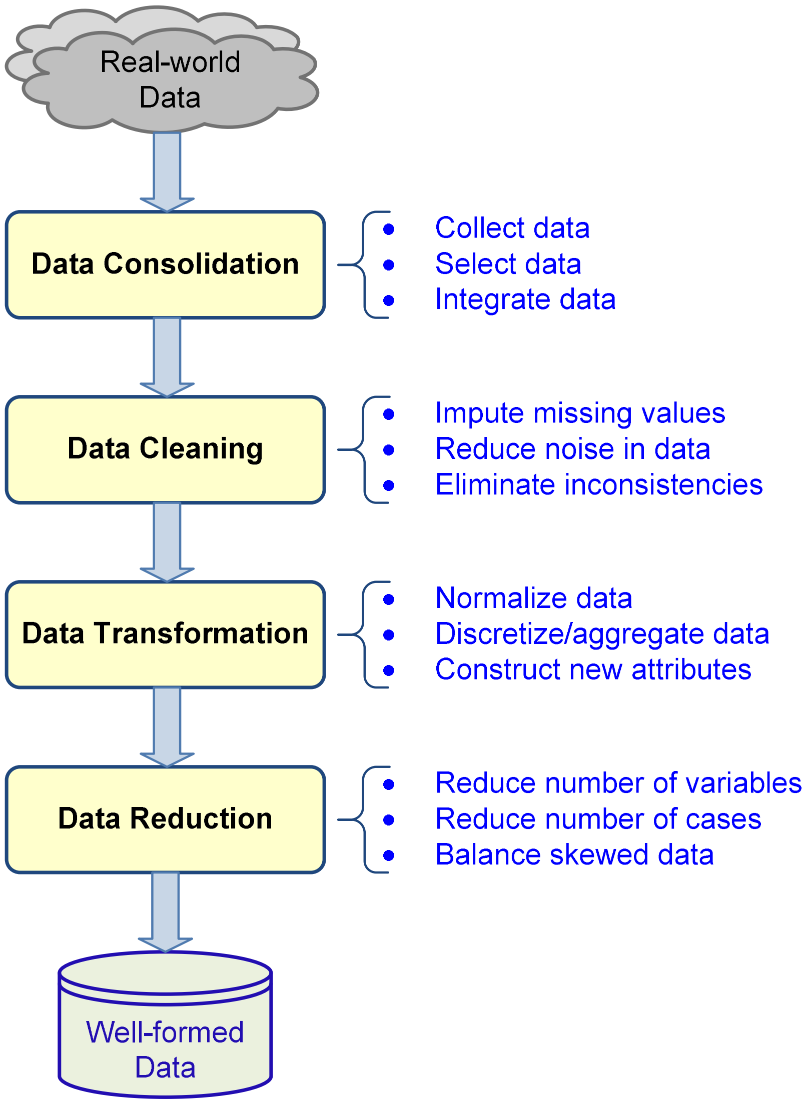
Step 5: Testing and Evaluation

Step 6: Deployment

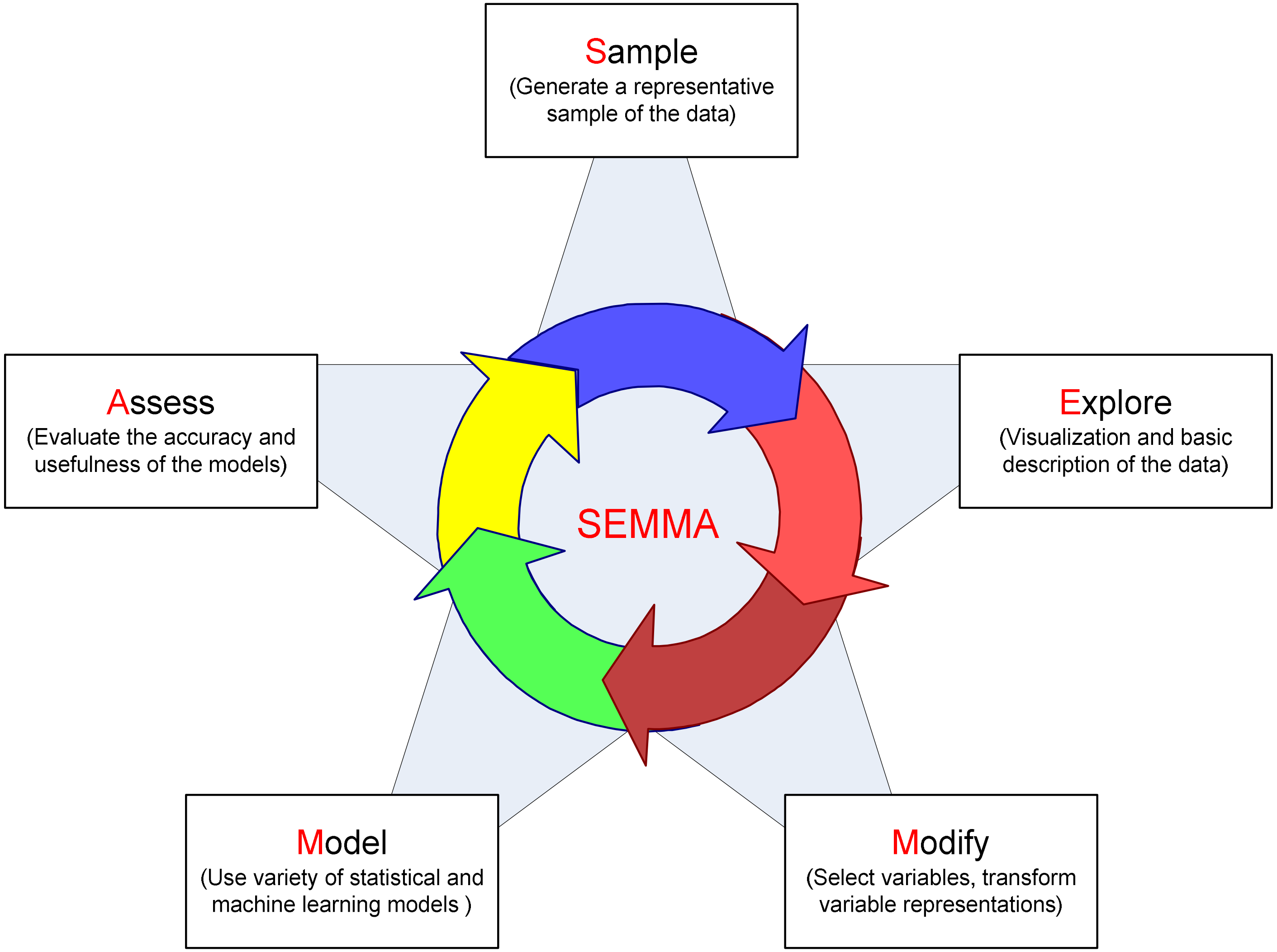
Accounts for ~85% of total project time

- The process is highly repetitive and experimental (DM: art versus science?)

Data Preparation – A Critical DM Task



Data Mining Process: SEMMA



Data Mining Methods: Classification

- Most frequently used DM method
- Part of the machine-learning family
- Employ supervised learning
- Learn from past data, classify new data
- The output variable is categorical (nominal or ordinal) in nature
- Classification versus regression?
- Classification versus clustering?

Assessment Methods for Classification

- Predictive accuracy
  - Hit rate
- Speed
  - Model building; predicting
- Robustness
- Scalability
- Interpretability
  - Transparency; ease of understanding

Accuracy of Classification Models

- In classification problems, the primary source for accuracy estimation is the confusion matrix

		True Class	
		Positive	Negative
Predicted Class	Positive	True Positive Count (TP)	False Positive Count (FP)
	Negative	False Negative Count (FN)	True Negative Count (TN)

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

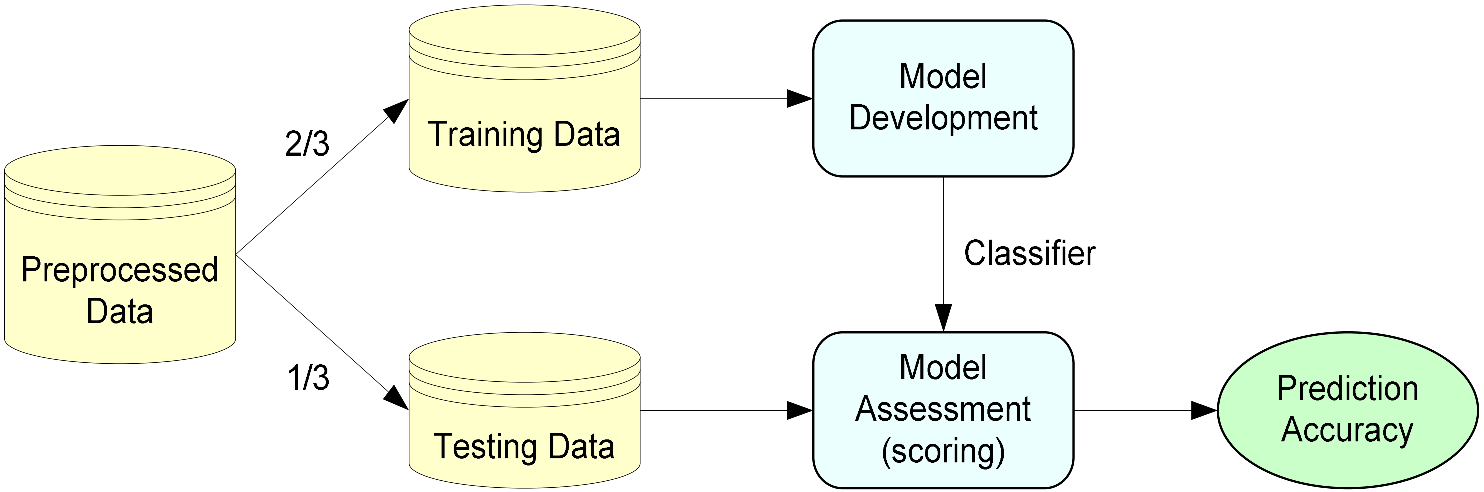
$$True\ Positive\ Rate = \frac{TP}{TP + FN}$$

$$True\ Negative\ Rate = \frac{TN}{TN + FP}$$

$$Precision = \frac{TP}{TP + FP} \qquad Recall = \frac{TP}{TP + FN}$$

Estimation Methodologies for Classification

- Simple split (or holdout or test sample estimation)
  - Split the data into 2 mutually exclusive sets training (~70%) and testing (30%)

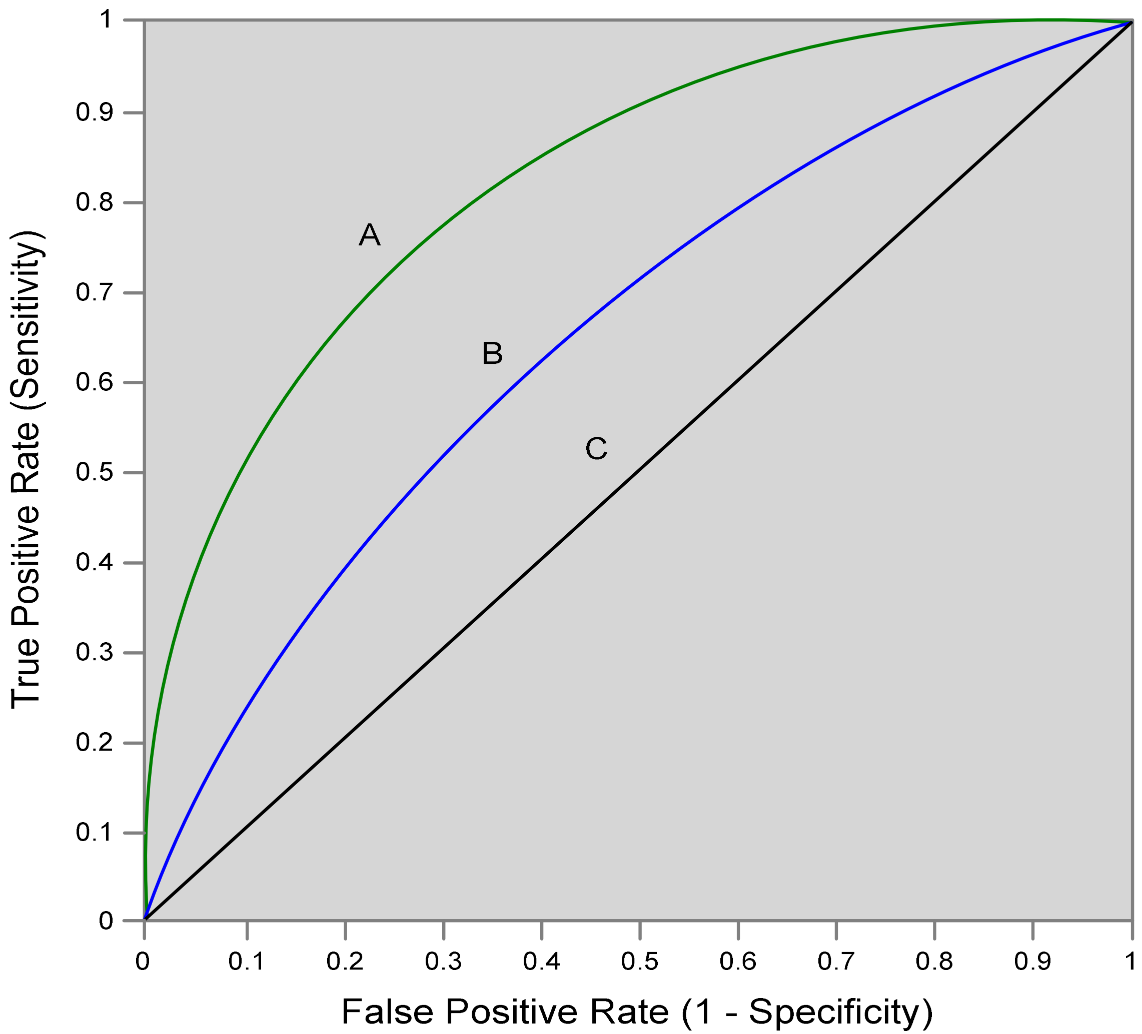


  - For ANN, the data is split into three sub-sets (training [~60%], validation [~20%], testing [~20%])

Estimation Methodologies for Classification

- *k*-Fold Cross Validation (rotation estimation)
  - Split the data into *k* mutually exclusive subsets
  - Use each subset as testing while using the rest of the subsets as training
  - Repeat the experimentation for *k* times
  - Aggregate the test results for true estimation of prediction accuracy training
- Other estimation methodologies
  - Leave-one-out, bootstrapping, jackknifing
  - Area under the ROC curve

Estimation Methodologies for Classification – ROC Curve



Classification Techniques

- Decision tree analysis
- Statistical analysis
- Neural networks
- Support vector machines
- Case-based reasoning
- Bayesian classifiers
- Genetic algorithms
- Rough sets

Decision Trees

- Employs the divide and conquer method
- Recursively divides a training set until each division consists of examples from one class

A general algorithm for decision tree building

Create a root node and assign all of the training data to it.

Select the best splitting attribute.

Add a branch to the root node for each value of the split. Split the data into mutually exclusive subsets along the lines of the specific split.

Repeat the steps 2 and 3 for each and every leaf node until the stopping criteria is reached.

Decision Trees

- DT algorithms mainly differ on
  - Splitting criteria
    - Which variable to split first?
    - What values to use to split?
    - How many splits to form for each node?
  - Stopping criteria
    - When to stop building the tree
  - Pruning (generalization method)
    - Pre-pruning versus post-pruning
- Most popular DT algorithms include
  - ID3, C4.5, C5; CART; CHAID; M5

Decision Trees

- Alternative splitting criteria
  - Gini index determines the purity of a specific class as a result of a decision to branch along a particular attribute/value
    - Used in CART
  - Information gain uses entropy to measure the extent of uncertainty or randomness of a particular attribute/value split
    - Used in ID3, C4.5, C5
  - Chi-square statistics (used in CHAID)

Cluster Analysis for Data Mining

- Used for automatic identification of natural groupings of things
- Part of the machine-learning family
- Employ unsupervised learning
- Learns the clusters of things from past data, then assigns new instances
- There is no output variable
- Also known as segmentation

Cluster Analysis for Data Mining

- Clustering results may be used to
  - Identify natural groupings of customers
  - Identify rules for assigning new cases to classes for targeting/diagnostic purposes
  - Provide characterization, definition, labeling of populations
  - Decrease the size and complexity of problems for other data mining methods
  - Identify outliers in a specific domain (e.g., rare-event detection)

Cluster Analysis for Data Mining

- Analysis methods
  - Statistical methods (including both hierarchical and nonhierarchical), such as *k*-means, *k*-modes, and so on.
  - Neural networks (adaptive resonance theory [ART], self-organizing map [SOM])
  - Fuzzy logic (e.g., fuzzy c-means algorithm)
  - Genetic algorithms
- Divisive versus Agglomerative methods

Cluster Analysis for Data Mining

- How many clusters?
  - There is no “truly optimal” way to calculate it
  - Heuristics are often used
    - Look at the sparseness of clusters
    - Number of clusters = $(n/2)^{1/2}$ (n: no of data points)
    - Use Akaike information criterion (AIC)
    - Use Bayesian information criterion (BIC)
- Most cluster analysis methods involve the use of a distance measure to calculate the closeness between pairs of items.
  - Euclidian versus Manhattan (rectilinear) distance

Cluster Analysis for Data Mining

- *k*-Means Clustering Algorithm
  - *k* : pre-determined number of clusters
  - Algorithm (Step 0: determine value of *k*)

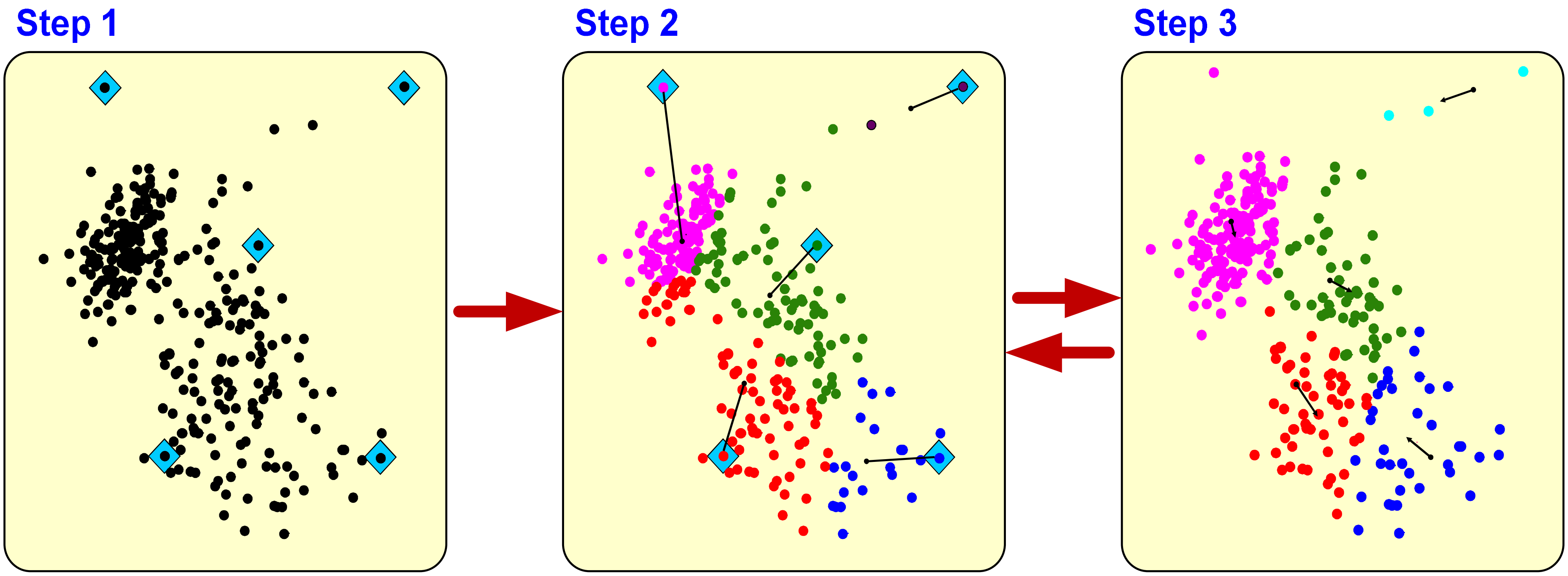
Step 1: Randomly generate *k* random points as initial cluster centers.

Step 2: Assign each point to the nearest cluster center.

Step 3: Re-compute the new cluster centers.

Repeat steps 3 and 4 until some convergence criterion is met (usually that the assignment of points to clusters becomes stable).

Cluster Analysis for Data Mining - *k*-Means Clustering Algorithm



Association Rule Mining

- A very popular DM method in business
- Finds interesting relationships (affinities) between variables (items or events)
- Part of machine learning family
- Employs unsupervised learning
- There is no output variable
- Also known as market basket analysis
- Often used as an example to describe DM to ordinary people, such as the famous "relationship between diapers and beers!"

Association Rule Mining

- Input: the simple point-of-sale transaction data
- Output: Most frequent affinities among items
- Example: according to the transaction data…

  "Customer who bought a laptop computer and a virus protection software, also bought extended service plan 70 percent of the time"
- How do you use such a pattern/knowledge?
  - Put the items next to each other for ease of finding
  - Promote the items as a package (do not put one on sale if the other(s) are on sale)
  - Place items far apart from each other so that the customer has to walk the aisles to search for it, and by doing so potentially see and buy other items

Association Rule Mining

- Representative applications of association rule mining include
  - In business: cross-marketing, cross-selling, store design, catalog design, e-commerce site design, optimization of online advertising, product pricing, and sales/promotion configuration
  - In medicine: relationships between symptoms and illnesses; diagnosis and patient characteristics and treatments (to be used in medical DSS); and genes and their functions (to be used in genomics projects)

Association Rule Mining

- Are all association rules interesting and useful?

A Generic Rule: **X $\Rightarrow$ Y [S%, C%]**

X, Y: products and/or services
X: Left-hand-side (LHS)
Y: Right-hand-side (RHS)
S: Support: how often **X** and **Y** go together
C: Confidence: how often **Y** go together with the **X**

Example: {Laptop Computer, Antivirus Software} $\Rightarrow$ {Extended Service Plan} [30%, 70%]

Association Rule Mining

- Algorithms are available for generating association rules
  - Apriori
  - Eclat
  - FP-Growth
  - + Derivatives and hybrids of the three
- The algorithms help identify the frequent item sets, which are, then converted to association rules

Association Rule Mining

- Apriori Algorithm
  - Finds subsets that are common to at least a minimum number of the itemsets
  - Uses a bottom-up approach
    - frequent subsets are extended one item at a time (the size of frequent subsets increases from one-item subsets to two-item subsets, then three-item subsets, and so on)
    - groups of candidates at each level are tested against the data for minimum support (*see the figure*) →

Association Rule Mining

- Apriori Algorithm

Raw Transaction Data

Transaction No	SKUs (Item No)
1	1, 2, 3, 4
1	2, 3, 4
1	2, 3
1	1, 2, 4
1	1, 2, 3, 4
1	2, 4

One-item Itemsets

Itemset (SKUs)	Support
1	3
2	6
3	4
4	5

Two-item Itemsets

Itemset (SKUs)	Support
1, 2	3
1, 3	2
1, 4	3
2, 3	4
2, 4	5
3, 4	3

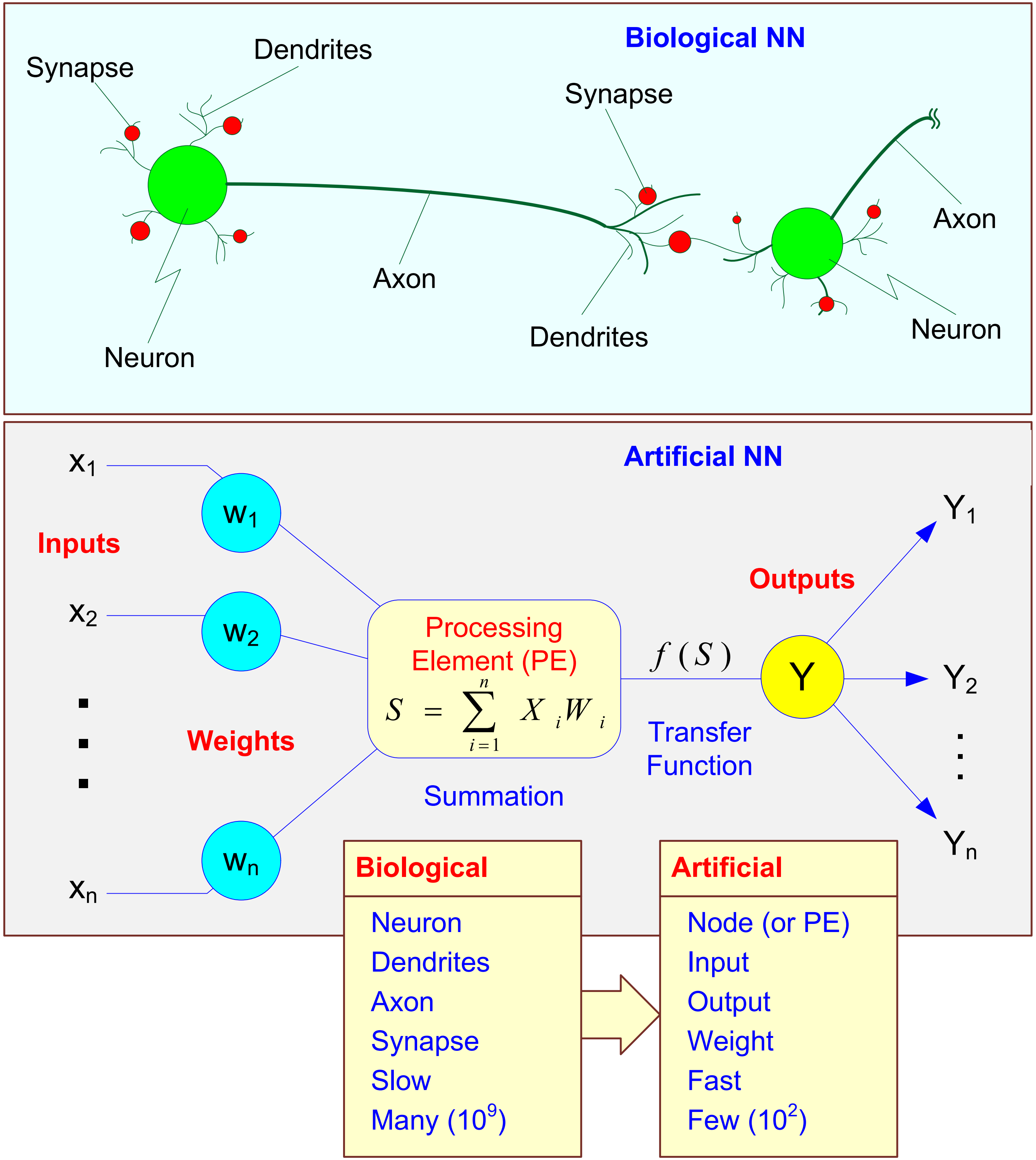
Three-item Itemsets

Itemset (SKUs)	Support
1, 2, 4	3
2, 3, 4	3

Artificial Neural Networks for Data Mining

- Artificial neural networks (ANN or NN) is a brain metaphor for information processing
- a.k.a. Neural Computing
- Very good at capturing highly complex non-linear functions!
- Many uses – prediction (regression, classification), clustering/segmentation
- Many application areas – finance, medicine, marketing, manufacturing, service operations, information systems, etc.

Biological versus Artificial Neural Networks

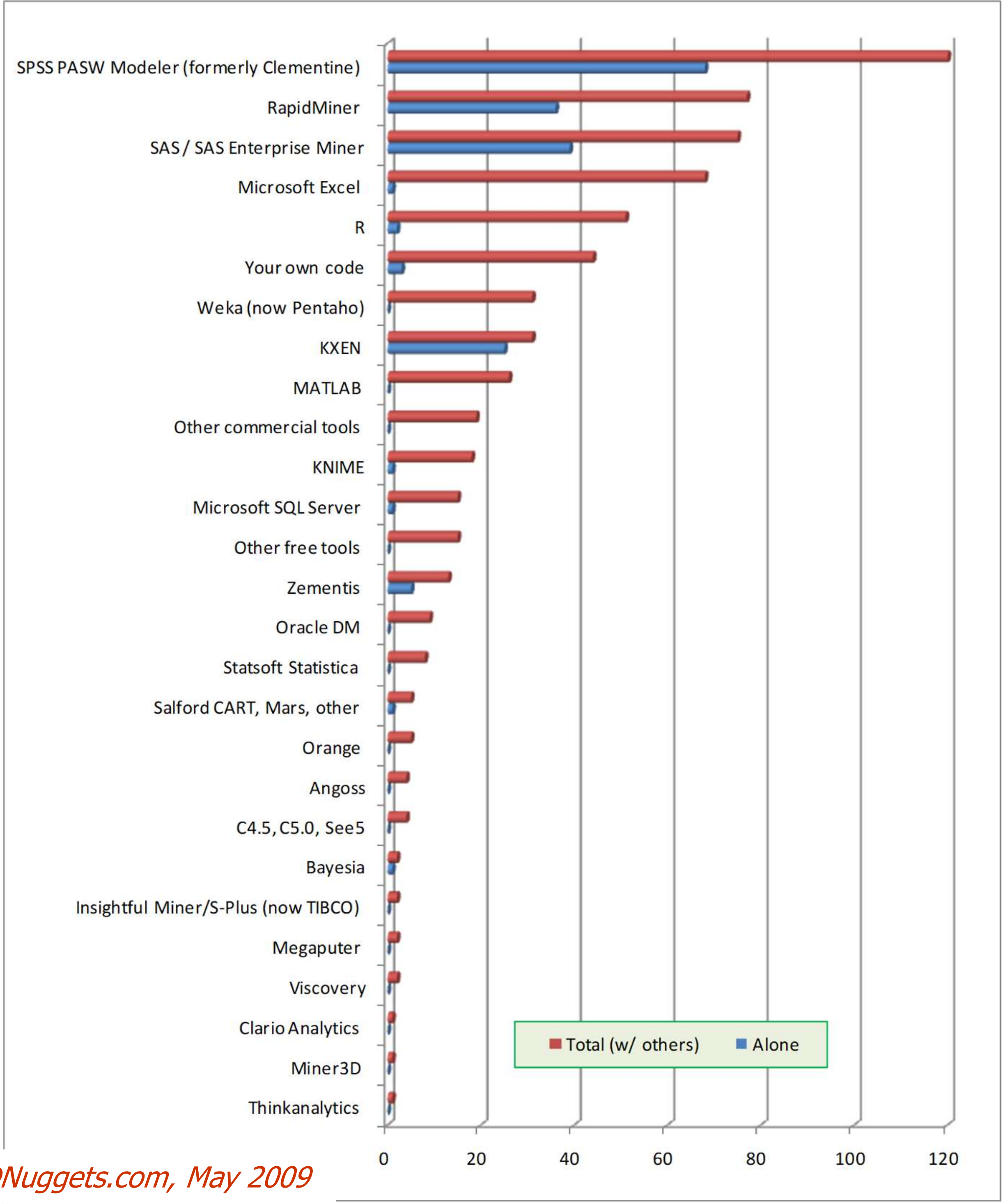


Elements/Concepts of ANN

- Processing element (PE)
- Information processing
- Network structure
  - Feedforward vs. recurrent vs. multi-layer…
- Learning parameters
  - Supervised/unsupervised, backpropagation, learning rate, momentum
- ANN Software – NN shells, integrated modules in comprehensive DM software, …

Data Mining Software

- Commercial
  - IBM SPSS Modeler (formerly Clementine)
  - SAS – Enterprise Miner
  - IBM – Intelligent Miner
  - StatSoft – Statistica Data Mine
  - … many more
- Free and/or Open Source
  - RapidMiner
  - Weka
  - … many more



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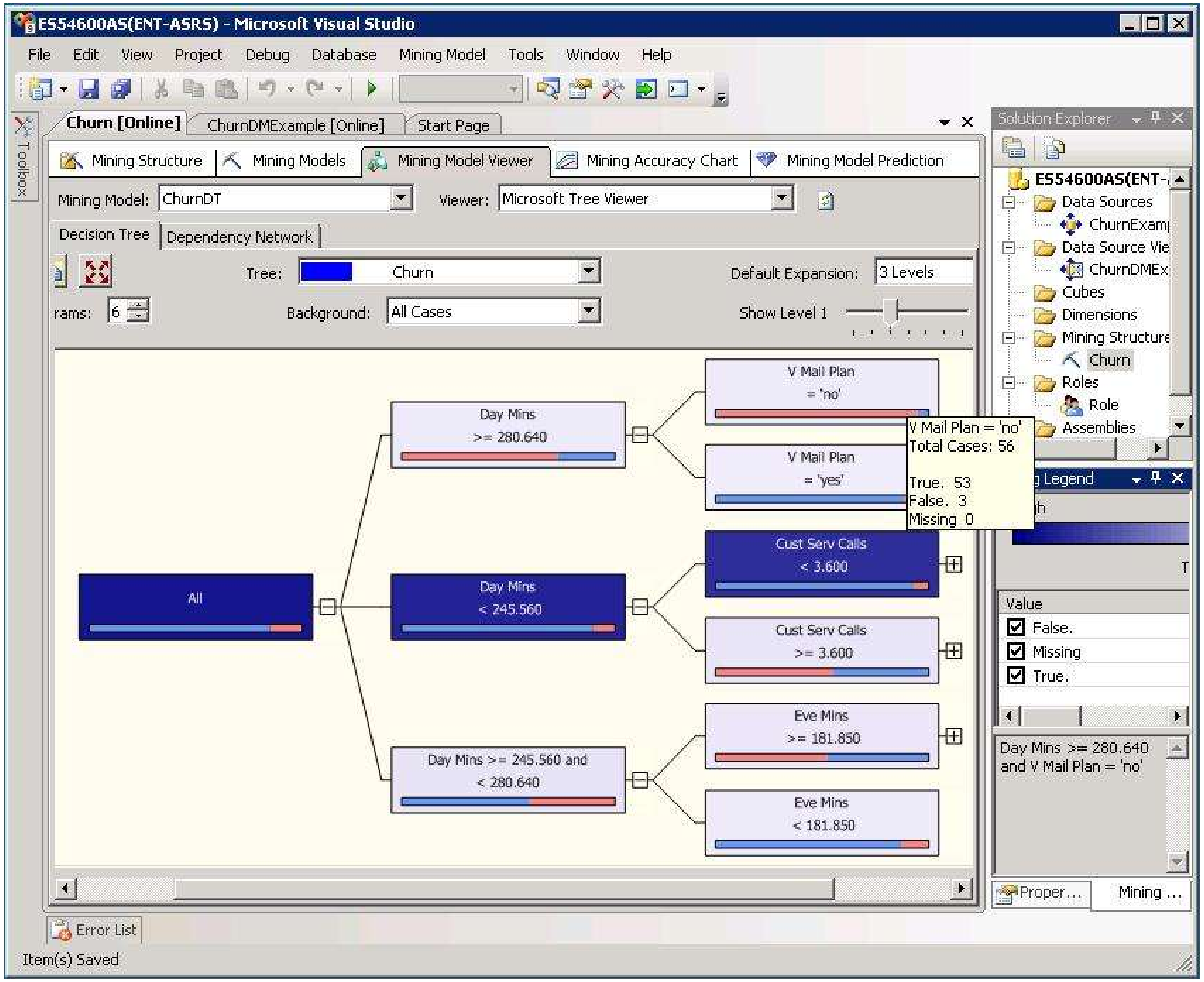
Data Mining in MS SQL Server 2008



Data Mining Myths

- Data mining …
  - provides instant solutions/predictions.
  - is not yet viable for business applications.
  - requires a separate, dedicated database.
  - can only be done by those with advanced degrees.
  - is only for large firms that have lots of customer data.
  - is another name for good-old statistics.

Common Data Mining Blunders

1. Selecting the wrong problem for data mining
2. Ignoring what your sponsor thinks data mining is and what it really can/cannot do
3. Not leaving sufficient time for data acquisition, selection and preparation
4. Looking only at aggregated results and not at individual records/predictions
5. Being sloppy about keeping track of the data mining procedure and results

Common Data Mining Mistakes

6. Ignoring suspicious (good or bad) findings and quickly moving on
7. Running mining algorithms repeatedly and blindly, without thinking about the next stage
8. Naively believing everything you are told about the data
9. Naively believing everything you are told about your own data mining analysis
10. Measuring your results differently from the way your sponsor measures them

End of the Chapter

- Questions, comments